PAPER PRESENTATION

Masked Autoencoders Are Scalable Vision Learners

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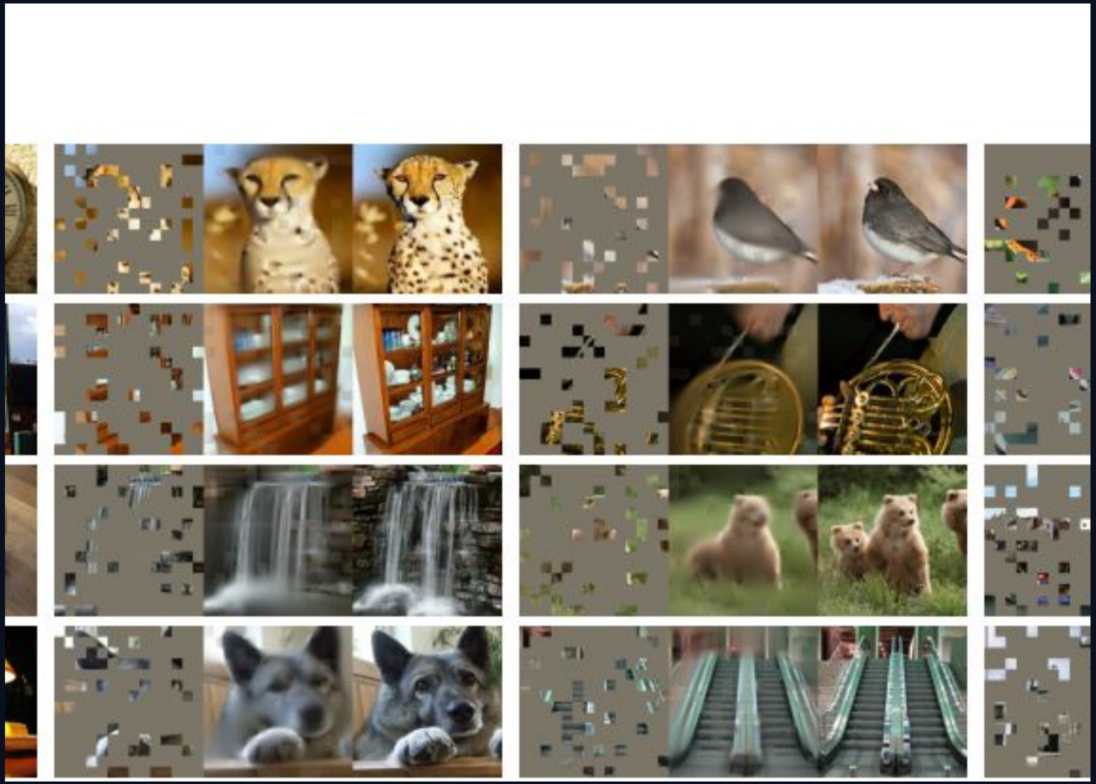
CVPR 2022 | Facebook AI Research (FAIR)



75%	3×+	87.8%
typical masked patches	training speedup	ViT-H ImageNet top-1

Why did this paper matter?

Large vision models need a lot of labeled data.

- Supervised learning depends on human labels.
- Bigger models can overfit when labels are limited.
- NLP succeeded with self-supervised pretraining: hide words, predict them.
- This paper asks: can we do a similar thing for images?

NLP idea
BERT masks words
→ model learns language

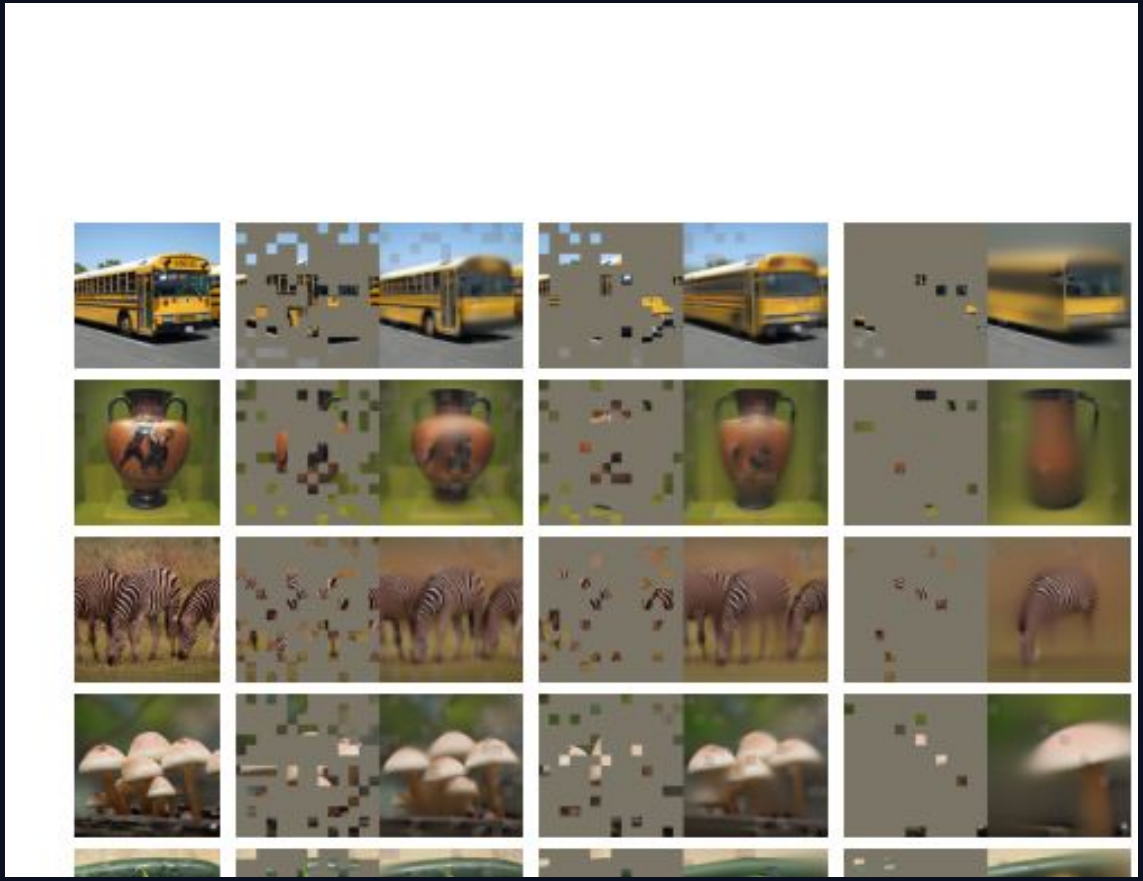
Vision idea
MAE masks image patches
→ model learns visual structure

Image → **Mask patches** → **Reconstruct**

Key question: how do we make masking hard enough for images?

MAIN IDEA

Core idea: learn by reconstructing missing patches

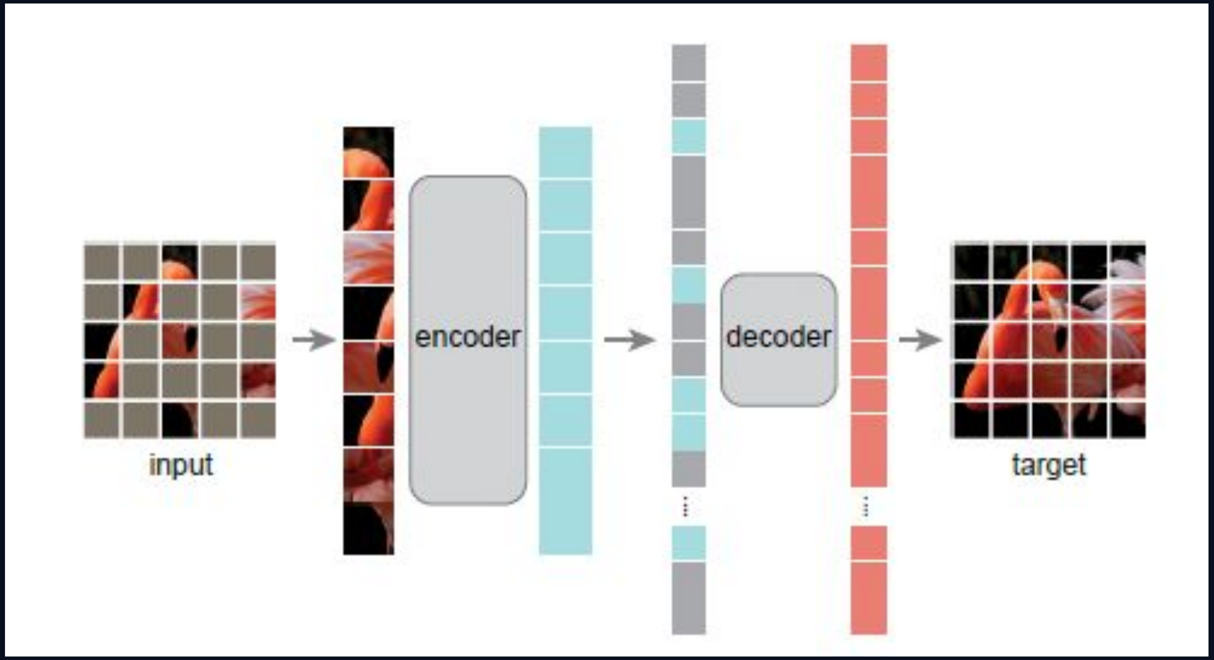


An MAE is a denoising-style autoencoder:

- Corruption = random patch masking.
- Input to the model = only the visible patches.
- Training target = the original image pixels.
- No human labels are needed.

This is self-supervised learning: the image creates its own label.

Architecture: asymmetric encoder-decoder



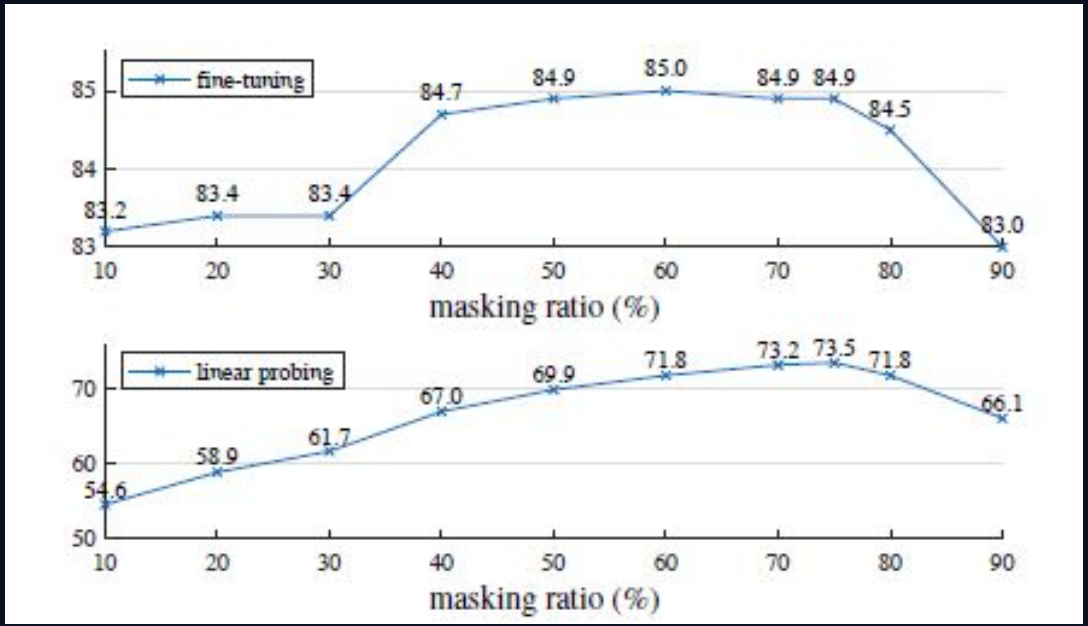
Two design choices make MAE efficient:

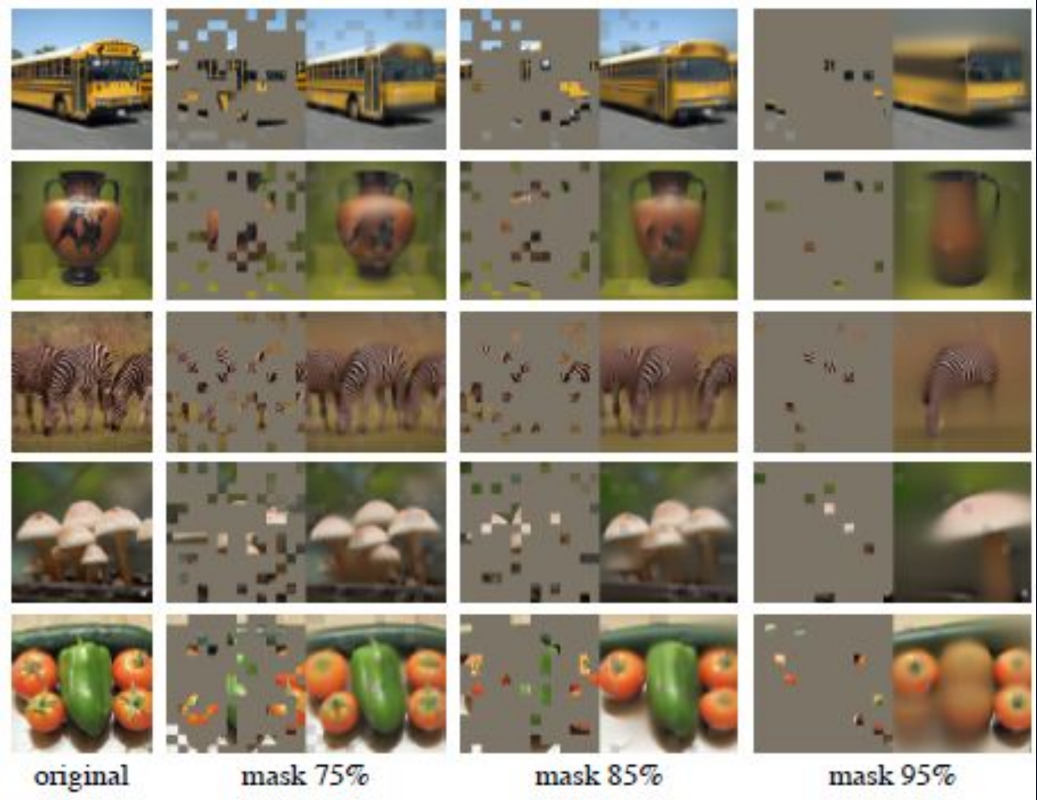
1. Encoder sees only visible patches
No mask tokens inside the encoder

2. Decoder is lightweight
It receives visible tokens + mask tokens

After pretraining, the decoder is discarded. The encoder is used for recognition tasks.

Why mask so much of the image?





- Images have high spatial redundancy.
- Low masking can be solved by local texture copying.
- High masking forces more global understanding.
- The paper finds 75% masking works very well.

Training objective and implementation

Split image into patches → Randomly keep 25% → Encode visible patches → Decode full image → Compute MSE loss

Loss is computed only on masked patches.

- Output target is pixel values of missing patches.
- They use Mean Squared Error (MSE).
- Normalized pixel targets improved representation quality.
- Implementation uses simple shuffle / remove / unshuffle operations.

Important intuition
The decoder solves reconstruction, but the encoder learns reusable visual features.

Ablation results: what actually mattered?

blocks	ft	lin
1	84.8	65.5
2	**84.9**	70.0
4	**84.9**	71.9
8	**84.9**	**73.5**
12	84.4	73.3

(a) **Decoder depth.** A deep decoder can improve linear probing accuracy.

dim	ft	lin
128	**84.9**	69.1
256	84.8	71.3
512	**84.9**	**73.5**
768	84.4	73.1
1024	84.3	73.1

(b) **Decoder width.** The decoder can be narrower than the encoder (1024-d).

case	ft	lin	FLOPs
encoder w/ [M]	84.2	59.6	3.3×
encoder w/o [M]	**84.9**	**73.5**	1×

(c) **Mask token.** An encoder without mask tokens is more accurate and faster (Table 2).

case	ft	lin
pixel (w/o norm)	84.9	73.5
pixel (w/ norm)	**85.4**	**73.9**
PCA	84.6	72.3
dVAE token	85.3	71.6

(d) **Reconstruction target.** Pixels as reconstruction targets are effective.

case	ft	lin
none	84.0	65.7
crop, fixed size	84.7	73.1
crop, rand size	**84.9**	**73.5**
crop + color jit	84.3	71.9

(e) **Data augmentation.** Our MAE works with minimal or no augmentation.

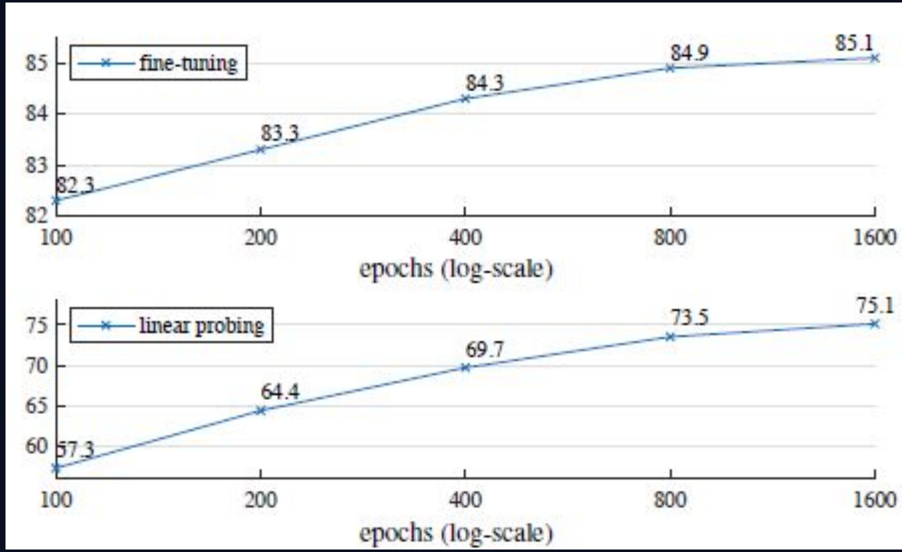
case	ratio	ft	lin
random	75	**84.9**	**73.5**
block	50	83.9	72.3
block	75	82.8	63.9
grid	75	84.0	66.0

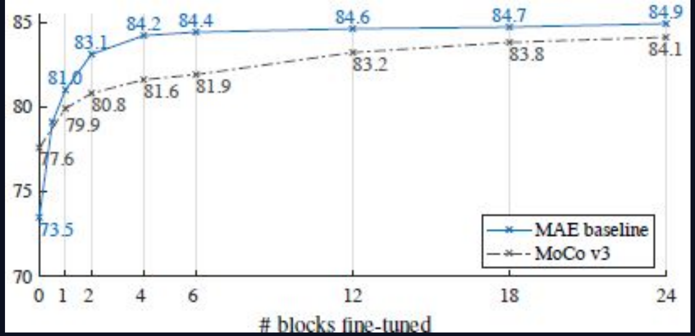
(f) **Mask sampling.** Random sampling works the best. See Figure 6 for visualizations.

Main lessons from the ablations:

- Removing mask tokens from the encoder is better and faster.
- A lightweight decoder is enough for fine-tuning.
- Random masking worked better than block or grid masking.
- MAE needs much less data augmentation than contrastive methods.
- Longer pretraining keeps improving performance.

Main result: MAE scales very well





83.6%
ViT-B ImageNet top-1

85.9%
ViT-L ImageNet top-1

87.8%
ViT-H at 448 resolution

- MAE improves as the model gets larger.
- It performs strongly using only ImageNet-1K data.
- It outperforms supervised pretraining in scaling behavior.
- Linear probing is not the full story; partial fine-tuning works much better.

Transfer learning: not only classification

method	pre-train data	ViT-B	ViT-L
supervised	IN1K w/ labels	47.4	49.9
MoCo v3	IN1K	47.3	49.1
BEiT	IN1K+DALLE	47.1	53.3
MAE	IN1K	**48.1**	**53.6**

Table 5. **ADE20K semantic segmentation** (mIoU) using Uper-

dataset	ViT-B	ViT-L	ViT-H	ViT-H$_{448}$	prev best
iNat 2017	70.5	75.7	79.3	**83.4**	75.4 [55]
iNat 2018	75.4	80.1	83.0	**86.8**	81.2 [54]
iNat 2019	80.5	83.4	85.7	**88.3**	84.1 [54]
Places205	63.9	65.8	65.9	**66.8**	66.0 [19]†
Places365	57.9	59.4	59.8	**60.3**	58.0 [40]‡

Table 6. **Transfer learning accuracy on classification datasets**,

	IN1K			COCO		ADE20K	
	ViT-B	ViT-L	ViT-H	ViT-B	ViT-L	ViT-B	ViT-L
pixel (w/o norm)	83.3	85.1	86.2	49.5	52.8	48.0	51.8
pixel (w/ norm)	83.6	85.9	86.9	50.3	53.3	48.1	53.6
dVAE token	83.6	85.7	86.9	50.3	53.2	48.1	53.4
△	0.0	-0.2	0.0	0.0	-0.1	0.0	-0.2

Table 7. **Pixels *vs.* tokens as the MAE reconstruction target.** △ is

They test whether MAE features transfer to other tasks.

COCO
object detection
+ instance
segmentation

ADE20K
semantic
segmentation

iNaturalist / Places
classification transfer

Result: MAE pretraining improves transfer performance, especially with larger models.

Strengths and limitations

Strengths

- Simple idea: mask patches and reconstruct pixels.
- Efficient: encoder processes only visible patches.
- Scales well to very large Vision Transformers.
- Strong transfer to detection and segmentation.
- Does not depend on labels during pretraining.

Limitations

- Masking is random, not physically meaningful noise.
- Still needs long pretraining schedules.
- Main experiments focus on natural images.
- Using ViT models may be computationally heavy.

Final takeaway

My understanding of the paper:

1
Hide most image patches

2
Train model to reconstruct pixels

3
Keep encoder as visual learner

Main message

A simple masked reconstruction task can train large vision models efficiently and produce strong features for many downstream tasks.